

RELATÓRIO DE DETECÇÃO DE INCIDENTES - SOC PREDITIVO

Análise de ML, validação de modelos e mapeamento MITRE ATT&CK;

■■ AVISO DE CONFIABILIDADE: a base analisada é pequena e/ou os modelos ainda não generalizam de forma estável (ver métricas na seção 4). Trate os números deste relatório como PROVA DE CONCEITO, não como validação estatística definitiva. Recomenda-se aumentar o volume de transações reais/rotuladas antes de usar estes scores operacionalmente.

1. Resumo Executivo

Este relatório documenta 225 transações sinalizadas por detecção de anomalias (Isolation Forest, contamination=0.15). O classificador de triagem atingiu ROC-AUC de 1.00 em dados de teste. O modelo de severidade (regressão) obteve R²=0.66 e erro médio absoluto de 14.0 pontos em uma escala de 0-100.

2. Detalhes Técnicos dos Alertas de Alta Severidade

Ref/ID	Cliente (LCPD)	Valor	Horário	Score Predito	Prob. Suspeita
1431	10000000-0000-4000-8000-000000000004	R\$ 19,403.92	6:00h	95.1/100	100%
1127	10000000-0000-4000-8000-000000000002	R\$ 70,676.12	23:00h	72.0/100	100%
1383	10000000-0000-4000-8000-000000000002	R\$ 63,829.78	1:00h	95.0/100	100%
1273	10000000-0000-4000-8000-000000000004	R\$ 88,144.26	7:00h	100.0/100	100%
1277	10000000-0000-4000-8000-000000000004	R\$ 48,410.80	8:00h	97.2/100	100%
1456	10000000-0000-4000-8000-000000000001	R\$ 58,055.64	10:00h	95.0/100	100%
580	10000000-0000-4000-8000-000000000005	R\$ 428.95	15:00h	7.7/100	0%
1488	10000000-0000-4000-8000-000000000003	R\$ 51,166.69	18:00h	67.7/100	100%

1223	10000000-0000-40 00-8000-00000000 0005	R\$ 59,055.64	7:00h	70.0/100	100%
1015	10000000-0000-40 00-8000-00000000 0003	R\$ 82,048.00	9:00h	99.3/100	100%
1215	10000000-0000-40 00-8000-00000000 0001	R\$ 40,361.70	19:00h	35.3/100	100%
1164	10000000-0000-40 00-8000-00000000 0004	R\$ 118,034.57	3:00h	100.0/100	100%
1120	10000000-0000-40 00-8000-00000000 0005	R\$ 61,328.31	13:00h	55.6/100	100%
1331	10000000-0000-40 00-8000-00000000 0003	R\$ 109,689.55	22:00h	95.9/100	100%
1009	10000000-0000-40 00-8000-00000000 0004	R\$ 111,250.69	2:00h	100.0/100	100%
1449	10000000-0000-40 00-8000-00000000 0002	R\$ 49,100.57	3:00h	81.8/100	100%
1040	10000000-0000-40 00-8000-00000000 0005	R\$ 44,505.99	22:00h	35.0/100	100%
1245	10000000-0000-40 00-8000-00000000 0003	R\$ 91,664.96	22:00h	69.6/100	100%
1258	10000000-0000-40 00-8000-00000000 0004	R\$ 85,198.09	0:00h	84.4/100	100%
1132	10000000-0000-40 00-8000-00000000 0004	R\$ 83,831.23	1:00h	82.2/100	100%
1293	10000000-0000-40 00-8000-00000000 0002	R\$ 104,030.83	3:00h	100.0/100	100%
1076	10000000-0000-40 00-8000-00000000 0002	R\$ 92,817.17	5:00h	98.6/100	100%
1178	10000000-0000-40 00-8000-00000000 0005	R\$ 57,555.32	7:00h	83.8/100	100%
1201	10000000-0000-40 00-8000-00000000 0005	R\$ 81,265.48	7:00h	87.6/100	100%

1385	10000000-0000-40 00-8000-00000000 0002	R\$ 61,493.28	13:00h	82.1/100	100%
1126	10000000-0000-40 00-8000-00000000 0002	R\$ 17,957.41	14:00h	49.2/100	100%
1187	10000000-0000-40 00-8000-00000000 0002	R\$ 85,716.20	17:00h	82.3/100	100%
1114	10000000-0000-40 00-8000-00000000 0002	R\$ 118,293.94	21:00h	88.9/100	100%
1292	10000000-0000-40 00-8000-00000000 0003	R\$ 89,433.31	0:00h	100.0/100	100%
1129	10000000-0000-40 00-8000-00000000 0004	R\$ 15,638.19	0:00h	48.9/100	100%
1375	10000000-0000-40 00-8000-00000000 0001	R\$ 62,710.61	6:00h	63.3/100	100%
1010	10000000-0000-40 00-8000-00000000 0003	R\$ 100,325.54	17:00h	94.2/100	100%
1185	10000000-0000-40 00-8000-00000000 0002	R\$ 70,438.53	22:00h	55.2/100	100%
1374	10000000-0000-40 00-8000-00000000 0003	R\$ 44,843.68	22:00h	36.0/100	100%
1443	10000000-0000-40 00-8000-00000000 0001	R\$ 56,337.96	23:00h	43.9/100	100%
1011	10000000-0000-40 00-8000-00000000 0002	R\$ 111,641.07	23:00h	97.0/100	100%
1042	10000000-0000-40 00-8000-00000000 0005	R\$ 72,939.83	2:00h	74.5/100	100%
1031	10000000-0000-40 00-8000-00000000 0005	R\$ 70,326.53	16:00h	86.7/100	100%
1106	10000000-0000-40 00-8000-00000000 0005	R\$ 15,926.11	16:00h	33.8/100	100%
1104	10000000-0000-40 00-8000-00000000 0004	R\$ 99,374.09	20:00h	78.6/100	100%

1336	10000000-0000-40 00-8000-00000000 0005	R\$ 114,386.78	20:00h	87.9/100	100%
1044	10000000-0000-40 00-8000-00000000 0004	R\$ 23,837.12	21:00h	36.1/100	100%
1256	10000000-0000-40 00-8000-00000000 0003	R\$ 65,041.89	23:00h	76.2/100	100%
1474	10000000-0000-40 00-8000-00000000 0002	R\$ 17,650.70	4:00h	58.2/100	100%
1319	10000000-0000-40 00-8000-00000000 0002	R\$ 24,199.24	6:00h	34.5/100	100%
1238	10000000-0000-40 00-8000-00000000 0002	R\$ 88,696.12	8:00h	100.0/100	100%
1430	10000000-0000-40 00-8000-00000000 0002	R\$ 20,378.76	9:00h	68.7/100	100%
1150	10000000-0000-40 00-8000-00000000 0001	R\$ 78,838.84	22:00h	87.0/100	100%
1146	10000000-0000-40 00-8000-00000000 0001	R\$ 76,322.66	0:00h	100.0/100	100%
1400	10000000-0000-40 00-8000-00000000 0005	R\$ 95,317.67	1:00h	91.0/100	100%
1033	10000000-0000-40 00-8000-00000000 0002	R\$ 59,518.91	6:00h	86.3/100	100%
1362	10000000-0000-40 00-8000-00000000 0001	R\$ 67,231.47	20:00h	67.2/100	100%
1141	10000000-0000-40 00-8000-00000000 0001	R\$ 19,464.43	21:00h	31.7/100	100%
1122	10000000-0000-40 00-8000-00000000 0003	R\$ 27,435.09	10:00h	73.2/100	100%
1196	10000000-0000-40 00-8000-00000000 0001	R\$ 119,596.13	16:00h	100.0/100	100%
1091	10000000-0000-40 00-8000-00000000 0003	R\$ 17,931.29	2:00h	46.3/100	100%

1099	10000000-0000-40 00-8000-00000000 0001	R\$ 111,285.59	23:00h	83.0/100	100%
1192	10000000-0000-40 00-8000-00000000 0004	R\$ 117,809.19	2:00h	100.0/100	100%
1410	10000000-0000-40 00-8000-00000000 0004	R\$ 22,119.56	22:00h	47.9/100	100%
1440	10000000-0000-40 00-8000-00000000 0002	R\$ 15,207.90	23:00h	40.5/100	100%
1206	10000000-0000-40 00-8000-00000000 0001	R\$ 49,277.60	23:00h	66.5/100	100%
1230	10000000-0000-40 00-8000-00000000 0002	R\$ 69,848.91	0:00h	100.0/100	100%
1500	10000000-0000-40 00-8000-00000000 0004	R\$ 43,339.21	14:00h	56.8/100	100%
1002	10000000-0000-40 00-8000-00000000 0004	R\$ 51,341.67	20:00h	70.4/100	100%
1034	10000000-0000-40 00-8000-00000000 0002	R\$ 89,871.56	23:00h	67.8/100	100%
1079	10000000-0000-40 00-8000-00000000 0004	R\$ 84,573.08	4:00h	100.0/100	100%
1415	10000000-0000-40 00-8000-00000000 0003	R\$ 101,760.55	5:00h	100.0/100	100%
1067	10000000-0000-40 00-8000-00000000 0001	R\$ 98,827.13	19:00h	78.5/100	100%
1307	10000000-0000-40 00-8000-00000000 0001	R\$ 31,358.03	21:00h	54.5/100	100%
1204	10000000-0000-40 00-8000-00000000 0001	R\$ 69,591.47	21:00h	82.2/100	100%
1119	10000000-0000-40 00-8000-00000000 0001	R\$ 18,597.73	23:00h	30.0/100	100%
1303	10000000-0000-40 00-8000-00000000 0001	R\$ 92,861.24	1:00h	89.8/100	100%

1355	10000000-0000-40 00-8000-00000000 0004	R\$ 118,549.42	22:00h	100.0/100	100%
1180	10000000-0000-40 00-8000-00000000 0004	R\$ 25,883.77	23:00h	35.2/100	100%
1262	10000000-0000-40 00-8000-00000000 0005	R\$ 62,917.72	23:00h	61.5/100	100%
1403	10000000-0000-40 00-8000-00000000 0004	R\$ 78,054.27	23:00h	86.3/100	100%
1190	10000000-0000-40 00-8000-00000000 0004	R\$ 80,582.90	1:00h	80.7/100	100%
1130	10000000-0000-40 00-8000-00000000 0001	R\$ 99,331.13	17:00h	93.5/100	100%
1380	10000000-0000-40 00-8000-00000000 0001	R\$ 74,014.63	23:00h	83.2/100	100%
1425	10000000-0000-40 00-8000-00000000 0002	R\$ 105,240.35	1:00h	97.7/100	100%
1144	10000000-0000-40 00-8000-00000000 0002	R\$ 39,643.11	2:00h	62.5/100	100%
1344	10000000-0000-40 00-8000-00000000 0003	R\$ 75,490.52	22:00h	58.0/100	100%
1427	10000000-0000-40 00-8000-00000000 0004	R\$ 64,089.60	2:00h	67.3/100	100%
1263	10000000-0000-40 00-8000-00000000 0002	R\$ 116,682.03	4:00h	100.0/100	100%
1470	10000000-0000-40 00-8000-00000000 0001	R\$ 84,864.69	22:00h	64.8/100	100%
1315	10000000-0000-40 00-8000-00000000 0001	R\$ 107,687.33	3:00h	100.0/100	100%
1438	10000000-0000-40 00-8000-00000000 0005	R\$ 70,564.10	3:00h	72.5/100	100%
1458	10000000-0000-40 00-8000-00000000 0003	R\$ 66,579.58	19:00h	69.0/100	100%

1240	10000000-0000-40 00-8000-00000000 0005	R\$ 98,803.03	20:00h	77.9/100	100%
1203	10000000-0000-40 00-8000-00000000 0005	R\$ 65,918.00	22:00h	78.9/100	100%
1259	10000000-0000-40 00-8000-00000000 0001	R\$ 35,647.69	23:00h	30.0/100	100%
1218	10000000-0000-40 00-8000-00000000 0003	R\$ 80,884.02	2:00h	94.0/100	100%
1121	10000000-0000-40 00-8000-00000000 0002	R\$ 15,284.12	5:00h	43.5/100	100%
1036	10000000-0000-40 00-8000-00000000 0002	R\$ 33,649.38	7:00h	68.3/100	100%
1353	10000000-0000-40 00-8000-00000000 0002	R\$ 63,601.50	7:00h	90.0/100	100%
1412	10000000-0000-40 00-8000-00000000 0002	R\$ 86,831.77	8:00h	100.0/100	100%
1296	10000000-0000-40 00-8000-00000000 0004	R\$ 68,235.47	23:00h	80.0/100	100%
1062	10000000-0000-40 00-8000-00000000 0003	R\$ 37,387.01	3:00h	87.9/100	100%
1020	10000000-0000-40 00-8000-00000000 0002	R\$ 94,968.73	22:00h	85.9/100	100%
1169	10000000-0000-40 00-8000-00000000 0005	R\$ 15,158.89	0:00h	60.8/100	100%
1194	10000000-0000-40 00-8000-00000000 0002	R\$ 92,933.39	1:00h	89.5/100	100%
1468	10000000-0000-40 00-8000-00000000 0003	R\$ 82,651.15	13:00h	98.3/100	100%
1006	10000000-0000-40 00-8000-00000000 0002	R\$ 89,829.78	18:00h	98.8/100	100%
1367	10000000-0000-40 00-8000-00000000 0005	R\$ 15,133.30	5:00h	69.5/100	100%

1489	10000000-0000-40 00-8000-00000000 0005	R\$ 21,665.58	5:00h	74.2/100	100%
1364	10000000-0000-40 00-8000-00000000 0003	R\$ 84,404.29	17:00h	82.4/100	100%
1107	10000000-0000-40 00-8000-00000000 0004	R\$ 86,291.89	0:00h	99.1/100	100%
1095	10000000-0000-40 00-8000-00000000 0001	R\$ 16,718.29	0:00h	61.7/100	100%
1097	10000000-0000-40 00-8000-00000000 0004	R\$ 39,903.13	3:00h	49.2/100	100%
1058	10000000-0000-40 00-8000-00000000 0004	R\$ 54,347.58	3:00h	59.6/100	100%
1086	10000000-0000-40 00-8000-00000000 0005	R\$ 15,096.96	23:00h	53.0/100	100%
1246	10000000-0000-40 00-8000-00000000 0005	R\$ 113,105.10	23:00h	100.0/100	100%
1183	10000000-0000-40 00-8000-00000000 0003	R\$ 88,838.05	1:00h	100.0/100	100%
1016	10000000-0000-40 00-8000-00000000 0004	R\$ 106,526.99	8:00h	94.6/100	100%
1461	10000000-0000-40 00-8000-00000000 0003	R\$ 95,280.13	18:00h	100.0/100	100%
1135	10000000-0000-40 00-8000-00000000 0003	R\$ 108,711.13	22:00h	96.7/100	100%
1384	10000000-0000-40 00-8000-00000000 0005	R\$ 35,385.11	23:00h	43.0/100	100%
1328	10000000-0000-40 00-8000-00000000 0005	R\$ 47,933.08	23:00h	52.1/100	100%
1266	10000000-0000-40 00-8000-00000000 0003	R\$ 79,152.61	1:00h	100.0/100	100%
1084	10000000-0000-40 00-8000-00000000 0004	R\$ 111,218.89	13:00h	93.4/100	100%

1484	10000000-0000-40 00-8000-00000000 0001	R\$ 110,032.11	17:00h	100.0/100	100%
1368	10000000-0000-40 00-8000-00000000 0004	R\$ 113,606.82	0:00h	100.0/100	100%
1411	10000000-0000-40 00-8000-00000000 0004	R\$ 47,100.57	2:00h	100.0/100	100%
1284	10000000-0000-40 00-8000-00000000 0005	R\$ 76,684.45	5:00h	100.0/100	100%
1056	10000000-0000-40 00-8000-00000000 0001	R\$ 22,934.75	7:00h	60.8/100	100%
1231	10000000-0000-40 00-8000-00000000 0001	R\$ 44,432.30	10:00h	73.7/100	100%
1311	10000000-0000-40 00-8000-00000000 0004	R\$ 16,796.12	4:00h	45.3/100	100%
1365	10000000-0000-40 00-8000-00000000 0004	R\$ 76,675.07	4:00h	88.8/100	100%
1413	10000000-0000-40 00-8000-00000000 0001	R\$ 97,910.52	7:00h	100.0/100	100%
1404	10000000-0000-40 00-8000-00000000 0005	R\$ 87,460.98	19:00h	83.1/100	100%
1314	10000000-0000-40 00-8000-00000000 0001	R\$ 92,124.24	8:00h	82.7/100	100%
1325	10000000-0000-40 00-8000-00000000 0005	R\$ 87,133.60	23:00h	65.4/100	100%
1499	10000000-0000-40 00-8000-00000000 0001	R\$ 23,386.36	22:00h	48.6/100	100%
1269	10000000-0000-40 00-8000-00000000 0004	R\$ 69,426.78	23:00h	80.9/100	100%
1069	10000000-0000-40 00-8000-00000000 0003	R\$ 54,205.45	23:00h	69.8/100	100%
1239	10000000-0000-40 00-8000-00000000 0004	R\$ 104,969.11	23:00h	100.0/100	100%

1177	10000000-0000-40 00-8000-00000000 0002	R\$ 117,722.45	20:00h	91.8/100	100%
1450	10000000-0000-40 00-8000-00000000 0005	R\$ 112,448.05	0:00h	100.0/100	100%
1165	10000000-0000-40 00-8000-00000000 0003	R\$ 73,490.25	2:00h	100.0/100	100%
1422	10000000-0000-40 00-8000-00000000 0003	R\$ 36,161.85	3:00h	86.9/100	100%
1053	10000000-0000-40 00-8000-00000000 0004	R\$ 93,886.44	21:00h	73.0/100	100%
1322	10000000-0000-40 00-8000-00000000 0003	R\$ 101,051.07	0:00h	100.0/100	100%
1112	10000000-0000-40 00-8000-00000000 0003	R\$ 18,021.57	1:00h	88.6/100	100%
1304	10000000-0000-40 00-8000-00000000 0003	R\$ 24,717.30	1:00h	93.5/100	100%
1394	10000000-0000-40 00-8000-00000000 0004	R\$ 115,251.87	3:00h	100.0/100	100%
1081	10000000-0000-40 00-8000-00000000 0005	R\$ 70,277.98	23:00h	93.7/100	100%
1387	10000000-0000-40 00-8000-00000000 0005	R\$ 100,166.45	8:00h	100.0/100	100%
1405	10000000-0000-40 00-8000-00000000 0004	R\$ 41,174.98	20:00h	90.1/100	100%
1419	10000000-0000-40 00-8000-00000000 0002	R\$ 89,862.25	1:00h	88.8/100	100%
1465	10000000-0000-40 00-8000-00000000 0001	R\$ 16,483.45	20:00h	32.4/100	100%
1471	10000000-0000-40 00-8000-00000000 0001	R\$ 100,177.76	20:00h	93.1/100	100%
1288	10000000-0000-40 00-8000-00000000 0001	R\$ 29,129.47	21:00h	40.7/100	100%

1308	10000000-0000-40 00-8000-00000000 0003	R\$ 54,564.23	2:00h	88.5/100	100%
1361	10000000-0000-40 00-8000-00000000 0004	R\$ 73,018.00	9:00h	95.7/100	100%
1268	10000000-0000-40 00-8000-00000000 0003	R\$ 29,117.46	1:00h	70.7/100	100%
1285	10000000-0000-40 00-8000-00000000 0004	R\$ 116,768.36	6:00h	100.0/100	100%
1092	10000000-0000-40 00-8000-00000000 0001	R\$ 37,282.22	23:00h	44.2/100	100%
1358	10000000-0000-40 00-8000-00000000 0003	R\$ 114,801.64	11:00h	100.0/100	100%
1101	10000000-0000-40 00-8000-00000000 0001	R\$ 20,358.28	12:00h	28.1/100	100%
1437	10000000-0000-40 00-8000-00000000 0003	R\$ 35,773.25	13:00h	64.6/100	100%
1212	10000000-0000-40 00-8000-00000000 0003	R\$ 56,980.03	23:00h	71.2/100	100%
1466	10000000-0000-40 00-8000-00000000 0001	R\$ 84,987.58	23:00h	64.9/100	100%
1398	10000000-0000-40 00-8000-00000000 0005	R\$ 84,219.71	12:00h	100.0/100	100%
1459	10000000-0000-40 00-8000-00000000 0002	R\$ 19,202.71	21:00h	45.2/100	100%
1464	10000000-0000-40 00-8000-00000000 0005	R\$ 23,677.82	0:00h	68.7/100	100%
1460	10000000-0000-40 00-8000-00000000 0005	R\$ 27,143.95	1:00h	83.6/100	100%
1085	10000000-0000-40 00-8000-00000000 0005	R\$ 34,562.56	1:00h	100.0/100	100%
1074	10000000-0000-40 00-8000-00000000 0002	R\$ 60,317.24	3:00h	92.7/100	100%

1396	10000000-0000-40 00-8000-00000000 0002	R\$ 88,851.86	9:00h	81.3/100	100%
1128	10000000-0000-40 00-8000-00000000 0002	R\$ 20,225.45	9:00h	44.8/100	100%
1406	10000000-0000-40 00-8000-00000000 0005	R\$ 89,150.88	18:00h	99.9/100	100%
1379	10000000-0000-40 00-8000-00000000 0003	R\$ 25,783.50	0:00h	69.7/100	100%
1281	10000000-0000-40 00-8000-00000000 0001	R\$ 90,142.81	1:00h	89.1/100	100%
1060	10000000-0000-40 00-8000-00000000 0003	R\$ 33,594.44	3:00h	72.6/100	100%
1234	10000000-0000-40 00-8000-00000000 0001	R\$ 119,358.79	5:00h	100.0/100	100%
1434	10000000-0000-40 00-8000-00000000 0003	R\$ 104,923.84	15:00h	100.0/100	100%
1208	10000000-0000-40 00-8000-00000000 0004	R\$ 16,753.76	23:00h	16.3/100	100%
1175	10000000-0000-40 00-8000-00000000 0003	R\$ 99,279.25	0:00h	96.1/100	100%
1480	10000000-0000-40 00-8000-00000000 0004	R\$ 81,445.11	4:00h	93.1/100	100%
1131	10000000-0000-40 00-8000-00000000 0005	R\$ 22,693.67	8:00h	73.2/100	100%
1280	10000000-0000-40 00-8000-00000000 0002	R\$ 101,168.55	19:00h	80.9/100	100%
1198	10000000-0000-40 00-8000-00000000 0004	R\$ 80,352.61	21:00h	77.2/100	100%
1179	10000000-0000-40 00-8000-00000000 0002	R\$ 15,502.50	22:00h	16.1/100	100%
1445	10000000-0000-40 00-8000-00000000 0003	R\$ 25,612.83	22:00h	49.8/100	100%

1054	10000000-0000-40 00-8000-00000000 0001	R\$ 78,189.68	23:00h	74.0/100	100%
1093	10000000-0000-40 00-8000-00000000 0004	R\$ 95,732.57	0:00h	100.0/100	100%
1134	10000000-0000-40 00-8000-00000000 0002	R\$ 82,436.47	3:00h	94.3/100	100%
1145	10000000-0000-40 00-8000-00000000 0003	R\$ 114,880.18	3:00h	100.0/100	100%
1158	10000000-0000-40 00-8000-00000000 0002	R\$ 29,370.10	4:00h	41.6/100	100%
1324	10000000-0000-40 00-8000-00000000 0004	R\$ 88,206.42	5:00h	83.2/100	100%
1357	10000000-0000-40 00-8000-00000000 0001	R\$ 114,444.33	7:00h	100.0/100	100%
1264	10000000-0000-40 00-8000-00000000 0003	R\$ 63,365.67	1:00h	95.1/100	100%
1457	10000000-0000-40 00-8000-00000000 0003	R\$ 55,813.45	2:00h	88.7/100	100%
1439	10000000-0000-40 00-8000-00000000 0001	R\$ 42,379.43	23:00h	60.6/100	100%
1409	10000000-0000-40 00-8000-00000000 0001	R\$ 42,191.84	0:00h	82.6/100	100%
1046	10000000-0000-40 00-8000-00000000 0004	R\$ 90,065.74	2:00h	88.7/100	100%
1216	10000000-0000-40 00-8000-00000000 0004	R\$ 61,921.84	3:00h	94.0/100	100%
1408	10000000-0000-40 00-8000-00000000 0002	R\$ 111,278.13	4:00h	100.0/100	100%
1237	10000000-0000-40 00-8000-00000000 0002	R\$ 53,239.32	5:00h	72.9/100	100%
1388	10000000-0000-40 00-8000-00000000 0004	R\$ 95,458.04	20:00h	76.7/100	100%

1300	10000000-0000-40 00-8000-00000000 0005	R\$ 94,058.81	22:00h	100.0/100	100%
1125	10000000-0000-40 00-8000-00000000 0004	R\$ 92,087.47	22:00h	99.1/100	100%
1366	10000000-0000-40 00-8000-00000000 0003	R\$ 18,982.70	2:00h	63.5/100	100%
1154	10000000-0000-40 00-8000-00000000 0003	R\$ 19,054.70	2:00h	76.8/100	100%
1105	10000000-0000-40 00-8000-00000000 0003	R\$ 25,088.75	3:00h	67.0/100	100%
1275	10000000-0000-40 00-8000-00000000 0002	R\$ 26,463.17	15:00h	57.7/100	100%
1075	10000000-0000-40 00-8000-00000000 0001	R\$ 69,030.85	16:00h	61.0/100	100%
1243	10000000-0000-40 00-8000-00000000 0001	R\$ 115,804.95	23:00h	100.0/100	100%
1205	10000000-0000-40 00-8000-00000000 0004	R\$ 102,466.60	4:00h	100.0/100	100%
1019	10000000-0000-40 00-8000-00000000 0003	R\$ 58,382.36	4:00h	90.0/100	100%
1478	10000000-0000-40 00-8000-00000000 0005	R\$ 98,754.31	7:00h	100.0/100	100%
1117	10000000-0000-40 00-8000-00000000 0001	R\$ 51,243.16	12:00h	78.0/100	100%
1428	10000000-0000-40 00-8000-00000000 0001	R\$ 58,515.38	19:00h	50.4/100	100%
1493	10000000-0000-40 00-8000-00000000 0003	R\$ 92,227.63	22:00h	72.0/100	100%
1381	10000000-0000-40 00-8000-00000000 0005	R\$ 16,317.40	23:00h	42.5/100	100%
1214	10000000-0000-40 00-8000-00000000 0003	R\$ 17,232.05	23:00h	29.9/100	100%

1301	10000000-0000-4000-8000-000000000002	R\$ 35,972.66	0:00h	50.8/100	100%
1219	10000000-0000-4000-8000-000000000005	R\$ 35,066.79	0:00h	76.2/100	100%
1078	10000000-0000-4000-8000-000000000003	R\$ 57,404.24	0:00h	100.0/100	100%
1014	10000000-0000-4000-8000-000000000004	R\$ 16,509.29	6:00h	57.7/100	100%
1272	10000000-0000-4000-8000-000000000002	R\$ 33,961.01	22:00h	56.4/100	100%
1157	10000000-0000-4000-8000-000000000003	R\$ 109,247.51	22:00h	100.0/100	100%
1108	10000000-0000-4000-8000-000000000003	R\$ 98,619.66	23:00h	100.0/100	100%
1343	10000000-0000-4000-8000-000000000001	R\$ 69,754.11	19:00h	58.0/100	100%
1155	10000000-0000-4000-8000-000000000003	R\$ 72,100.97	21:00h	57.7/100	100%

3. Correlação de Threat Intelligence (MITRE ATT&CK;)

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** múltiplas falhas de login antes da transação (força bruta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-0000000000004 às 6:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-0000000000002 às 23:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsvlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** múltiplas falhas de login antes da transação (força bruta).

- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000002 às 1:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** múltiplas falhas de login antes da transação (força bruta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000001 às 10:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000005 às 15:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** múltiplas falhas de login antes da transação (força bruta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000003 às 18:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000005 às 7:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsvlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000003 às 22:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsvlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000004 às 2:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsrvlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000002 às 21:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

Técnica associada: Valid Accounts (ID: **T1078**), tática **Stealth**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** mudança de localização entre acessos recentes (uso de credencial fora do padrão).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000002 às 6:00h.
- **Procedimentos sugeridos:** During RedPenguin, UNC3886 used legitimate credentials to gain privileged access to Juniper routers.(Citation: Mand... | During Operation MidnightEclipse, threat actors extracted sensitive credentials while moving laterally through compromised networks.(Citation: Volexity UPST... | During the 2015 Ukraine Electric Power Attack, Sandworm Team used valid accounts on the corporate network to escala...

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000005 às 1:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000001 às 20:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsrvlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000001 às 23:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

Técnica associada: Valid Accounts (ID: **T1078**), tática **Stealth**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** mudança de localização entre acessos recentes (uso de credencial fora do padrão).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-0000000000005 às 3:00h.
- **Procedimentos sugeridos:** During RedPenguin, UNC3886 used legitimate credentials to gain privileged access to Juniper routers.(Citation: Mand... | During Operation MidnightEclipse, threat actors extracted sensitive credentials while moving laterally through compromised networks.(Citation: Volexity UPST... | During the 2015 Ukraine Electric Power Attack, Sandworm Team used valid accounts on the corporate network to escala...

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-0000000000003 às 22:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

4. Validação dos Modelos (transparência metodológica)

Amostra: 1125 transações de treino / 375 de teste. Classificador de triagem: precision=1.00, recall=1.00 (reproduz decisões históricas, não detecta ataques inéditos). Detector selecionado (isolation forest): precision=1.00, recall=0.45, F1=0.62 (rótulo usado somente para auditoria comparativa). Regressão de severidade: R² (validação cruzada 5-fold)=-163354.27.

5. Comparação dos Detectores de Anomalia

Modelo	Alertas	Precision	Recall	F1	Tempo (s)
isolation forest	225	1.00	0.45	0.62	0.234
elliptic envelope	225	1.00	0.45	0.62	0.093
one class svm	228	0.88	0.40	0.55	0.039
local outlier factor	209	0.60	0.25	0.35	0.046

O detector selecionado representa o melhor desempenho obtido entre os modelos avaliados neste conjunto de dados, considerando F1-score, recall, precision e tempo de execução. Como a avaliação foi realizada sobre uma base sintética, os resultados demonstram a viabilidade da abordagem proposta e podem servir como referência para estudos futuros utilizando dados reais.